Data-driven SPS/DPS separation

Prompt-purified cross-slot mixing and current uncertainties

Speaker: Lekai Yao

7 August 2026

Can Data provide an independent SPS/DPS separation?

Why revisit the method?

- CMS nominal uses MC templates
- SPS modeling remains generator-dependent
- Data offer an independent decomposition

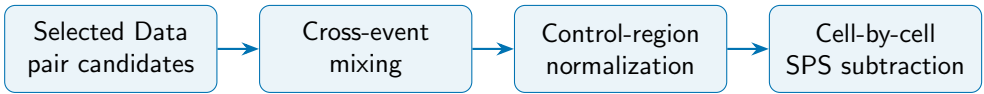
This update

- Purify prompt single-J/ψ sources
- Remove random-pair sampling
- Propagate full-chain Data statistics

Question: how stable is the data-driven fraction after these upgrades?

ATLAS Collaboration, Eur. Phys. J. C 77 (2017) 76, arXiv:1612.02950

Recap: the data-driven decomposition



DPS factorization assumption

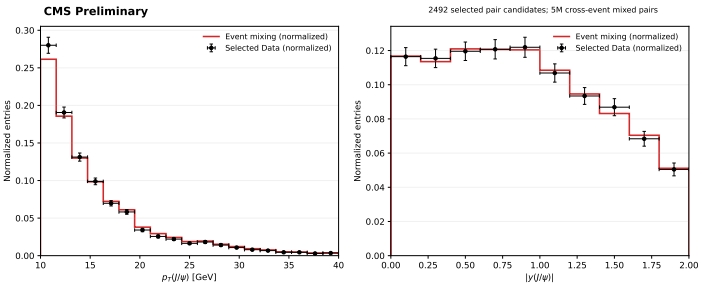
$$P_{\mathrm{DPS}}(J/\psi_1, J/\psi_2) \simeq P(J/\psi_1)P(J/\psi_2).$$

Data-driven subtraction

$$N^i_{\mathrm{SPS}} = N^i_{\mathrm{PP}} - \alpha N^i_{\mathrm{mix}}.$$

The method constrains shapes and the integrated fraction in the same 14 cells.

Recap: what had already been validated?



Previous accepted checks

- Reselected mixed-pair kinematics
- Stable inclusive single-J/ψ marginals
- Common adaptive 14-cell grid
- PP fit and algebraic closure

Open at that stage

The source pool was not prompt-purified before mixing.

Recap: validated cross-check, unresolved assumptions

Element	Established	Remaining question
PP extraction	Stable 4D fits	Model dependence
Mixed DPS	Cross-event construction	Prompt-source purity
Normalization	Reproducible CR scaling	DPS dominance of CR
Uncertainty	Statistical machinery	End-to-end covariance

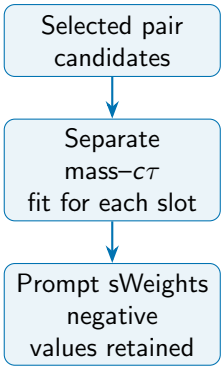
Role

Independent data-driven cross-check of the SPS/DPS decomposition.

Boundary

Not a replacement for the CMS MC-template nominal.

Upgrade 1: prompt-purify each single-J/ψ slot



Data input

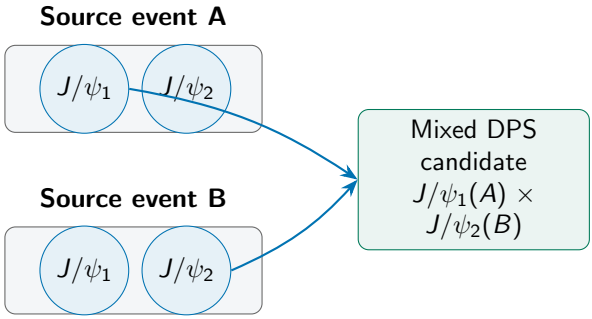
- 119 raw files
- 2492 selected events
- 2473 / 2474 fitted candidates

Fit content

- Prompt, nonprompt, combinatorial
- Slot-specific mass and lifetime response
- Yield-only sPlot after calibration

Prompt purification now occurs before the DPS template is built.

Upgrade 2: preserve the two reconstructed slots



Retained

- Slot-specific sWeights
- Signed pair weights
- Nominal pair correction

Rejected

- Same-source combinations
- $m(J/\psi J/\psi) < 7.5$ GeV
- Pairs failing correction inputs

Upgrade 3: deterministic all-pairs mixing

Pair accounting	Count
Cartesian $J/\psi_1 \times J/\psi_2$ combinations	6,118,202
Same-source rejected	2,456
$m(J/\psi J/\psi) < 7.5$ GeV rejected	53,997
Correction rejected	0
Accepted mixed pairs	**6,061,749**

Removed

Random-pair seed and finite sampling variance.

Retained

Every accepted cross-event pair with its signed weight.

All subsequent category, CR and bootstrap studies use the same deterministic construction.

Upgrade 4: calibrate the visible kinematic dependence

Single-J/ψ treatment	Split	Direct f_{DPS}	Role
Global	none	0.176303	variation
p_{T} categories	15 GeV	0.177244	variation
$\lvert y\rvert$ categories	1.2	**0.169980**	**central**

For each slot and category

- Calibrate mass resolution
- Calibrate prompt-$c\tau$ resolution
- Freeze scales; refit yields

Fit validation

Eight calibrations and eight yield fits pass; both category schemes complete all component fits.

The $|y| = 1.2$ split captures a consistent forward response change in both slots.

Current data-driven central definition

Inputs and estimator

- Current CMS pair selection
- $|y| = 1.2$ slot-specific prompt sPlots
- Cross-slot deterministic all-pairs
- Adaptive 14-cell PP fit

Control region

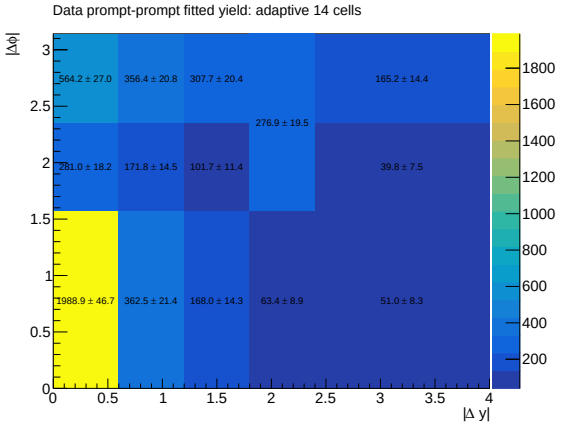
$$|\Delta y| \geq 1.8, \qquad |\Delta\phi| \leq \pi/2.$$

Integrated estimator

$$f_{\mathrm{DPS}} = \frac{\sum_i N^i_{\mathrm{DPS}}}{\sum_i N^i_{\mathrm{PP}}} = \mathbf{0.169980}.$$

Internal data-driven central; the CMS MC-template result remains the analysis nominal.

Central input: prompt–prompt Data yields



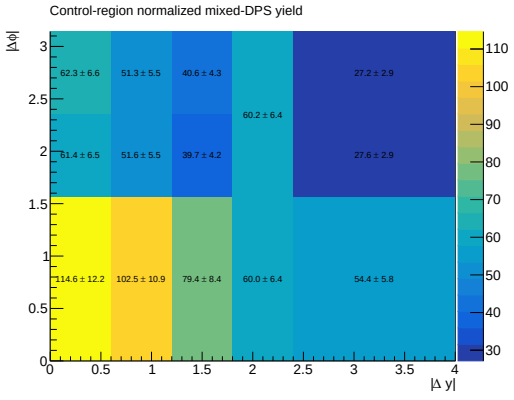
Four-dimensional fits

- 14 adaptive cells
- Mass and lifetime of both J/ψ candidates
- Prompt–prompt yield per cell
- All cell fits accepted

Role

Fitted Data input to the decomposition, not an independent fraction measurement.

Central component: normalized data-derived DPS



Control-region scale

$\alpha = N_{\mathrm{PP}}^{\mathrm{CR}} / N_{\mathrm{mix}}^{\mathrm{CR}}$

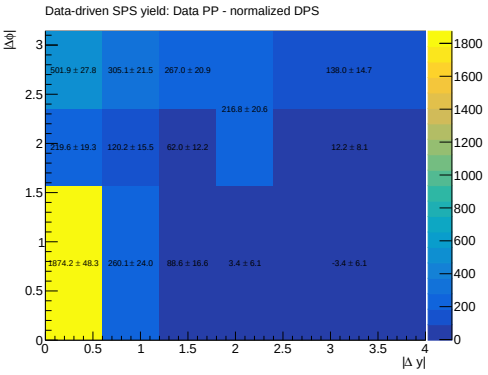
$N_{\mathrm{DPS}}^{i} = \alpha N_{\mathrm{mix}}^{i}$

- Signed mixed-pair weights retained
- Normalization covariance included
- Same 14-cell geometry as Data

Interpretation

CR equality is imposed by normalization; it is not a DPS-purity proof.

Central component: data-driven SPS by subtraction



Cell estimator

$$N^i_{\mathrm{SPS}} = N^i_{\mathrm{PP}} - N^i_{\mathrm{DPS}}.$$

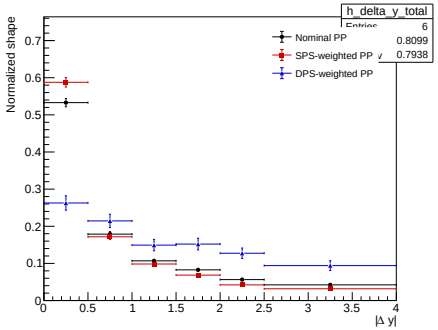
- Exact algebraic closure
- One signed negative SPS cell
- Compatible with zero; no clipping

Integrated result

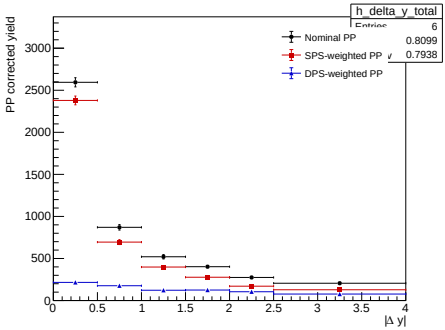
$f_{\mathrm{SPS}} = 0.830020$, hence
$f_{\mathrm{DPS}} = 0.169980$.

Component projections: rapidity separation

Unit-area shapes



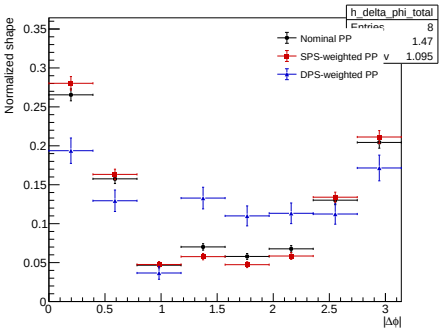
PP-corrected yields



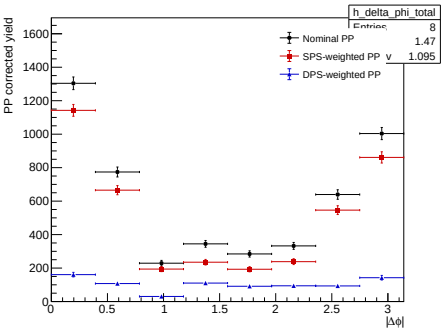
$f_{\mathrm{DPS}}^{|\Delta y|} = 0.168704$ – **projection consistency, not an independent measurement**

Component projections: azimuthal separation

Unit-area shapes



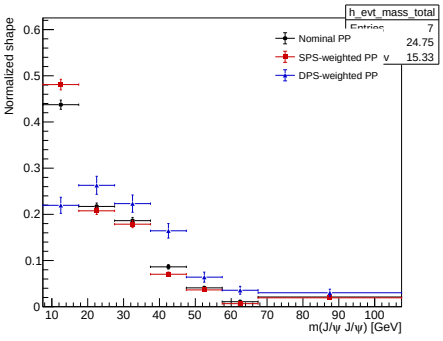
PP-corrected yields



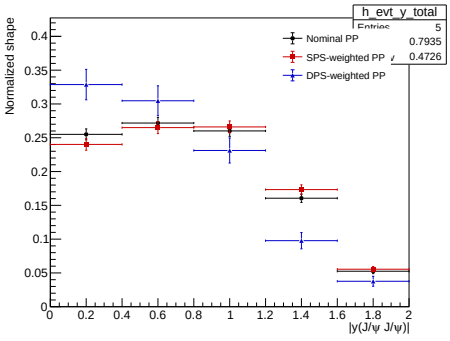
$f_{\mathrm{DPS}}^{|\Delta\phi|} = 0.169247$ – **same two-dimensional decomposition and correlations**

Component projections: pair mass and rapidity

$m(J/\psi J/\psi)$ **shapes**



$|y(J/\psi J/\psi)|$ **shapes**



$f^{m}_{\mathrm{DPS}} = 0.168262, \quad f^{|y|}_{\mathrm{DPS}} = 0.168745$ – all 78 component fits accepted

Category dependence: the largest positive variation

Scheme	f_{DPS}	Shift from central	Fit status
$\lvert y\rvert = 1.2$ categories	**0.169980**	—	all pass
Global response	0.176303	+0.006322	all pass
$p_{\mathrm{T}} = 15$ GeV categories	0.177244	+0.007264	all pass

Source tested

Single-J/ψ kinematic response and sPlot category choice.

Category envelope

$$\Delta f_{\mathrm{DPS}}^{\mathrm{cat}} = {}^{+0.007264}_{-0.000000}.$$

The variation changes the single-J/ψ calibration, not the pair selection or 14-cell estimator.

Control-region boundary sensitivity

Control-region variation	f_{DPS}	Signed shift	Fits
$\lvert\Delta\phi\rvert \leq 70^\circ$	0.171299	$+0.001319$	78/78
$\lvert\Delta\phi\rvert \leq 110^\circ$	0.167433	-0.002548	78/78
$\lvert\Delta y\rvert \geq 2.4$	0.159390	-0.010591	78/78

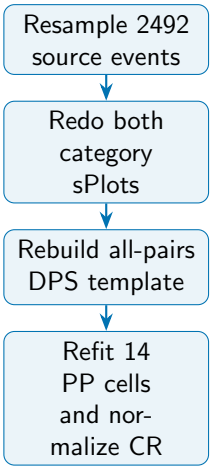
CR envelope

$$\Delta f_{\mathrm{DPS}}^{\mathrm{CR}} = {}^{+0.001319}_{-0.010591}.$$

Scope

Boundary stability only; neither a CR-purity measurement nor an event-mixing uncertainty.

Full-chain bootstrap statistical uncertainty



Summary	Value
Successful replicas	500
Attempts	529
Mean	0.164043
Median	0.159379
Sample SD	**0.056796**
16% / 84%	0.107694 / 0.223979

Successful-sample SD is labeled ‘stat’.

29 failures: 14/7 slot calibrations; 8 sparse-cell fits.

Provisional systematic: keep the sources visible

Source group	Up	Down	Interpretation
Category choice	$+0.007264$	-0.000000	single-J/ψ response
CR boundaries	$+0.001319$	-0.010591	normalization stability
Quadrature	$+0.007382$	-0.010591	**provisional syst**

$$\delta_{\mathrm{up}} = \sqrt{0.007264^2 + 0.001319^2}, \qquad \delta_{\mathrm{down}} = \sqrt{0^2 + 0.010591^2}.$$

Current result

$$f_{\mathrm{DPS}} = 0.169980 \pm 0.056796\ (\mathrm{stat.})\ {}^{+0.007382}_{-0.010591}\ (\mathrm{syst.})$$

Still missing

Alternative PDFs, correction uncertainties, binning, CR purity and factorization bias.

Results: data-driven cross-check and CMS nominal

Data-driven cross-check

Prompt-purified cross-slot mixing

$$f_{\mathrm{DPS}} = 0.170 \pm 0.057\,(\mathrm{stat.})^{+0.007}_{-0.011}\,(\mathrm{prov.\ syst.})$$

$$\sigma_{\mathrm{eff}} = 2.51^{+0.26}_{-0.20}\,(\mathrm{partial\ syst.})\ \mathrm{mb}$$

Statistical uncertainty pending covariance.

CMS MC-template nominal

Simultaneous SPS/DPS template fit

$$f_{\mathrm{DPS}} = 0.34 \pm 0.10\,(\mathrm{stat.})^{+0.02}_{-0.17}\,(\mathrm{syst.})$$

$$\sigma_{\mathrm{eff}} = 1.26 \pm 0.37\,(\mathrm{stat.})^{+0.64}_{-0.11}\,(\mathrm{syst.})\ \mathrm{mb}$$

Current analysis nominal.

Different methods and systematic coverage: compare the quoted results, not individual uncertainty components.

Takeaways and current status

Status	Established now	Remaining limitation
Source candidates	Prompt-purified per slot	Fixed-shape model dependence
Mixed template	Deterministic all-pairs	SPS/DPS marginal factorization
Statistics	500 full-chain successes	29 conditional failures
Systematics	Category + CR groups	Incomplete source coverage

1. **Method upgrade:** prompt purification, slot separation and deterministic mixing.
2. **Current cross-check:** $f_{\mathrm{DPS}} = 0.169980 \pm 0.056796\,(\mathrm{stat.})^{+0.007382}_{-0.010591}\,(\mathrm{syst.})$.
3. **Next priority:** complete model, correction, binning and CR-purity coverage.

The data-driven chain is now end-to-end, but its systematic program is not yet complete.

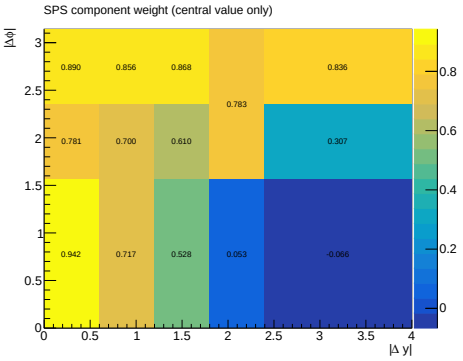
Thanks for listening!

Reporter: Lekai Yao

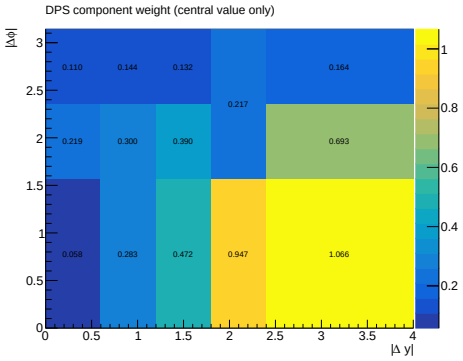
7 August 2026

Backup: central component weights

SPS weights

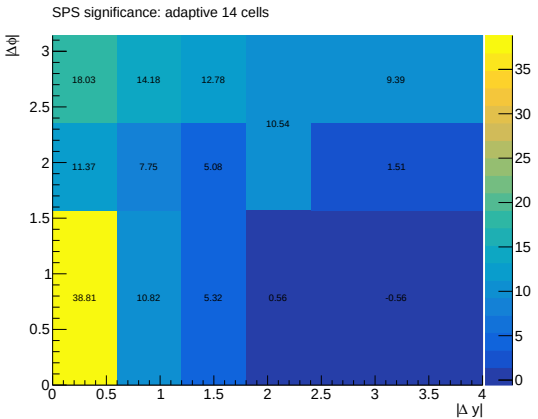


DPS weights



$w^i_{\mathrm{SPS}} + w^i_{\mathrm{DPS}} = 1$ – central values only; full ratio covariance is not encoded

Backup: signed-cell diagnostic



Diagnostic only

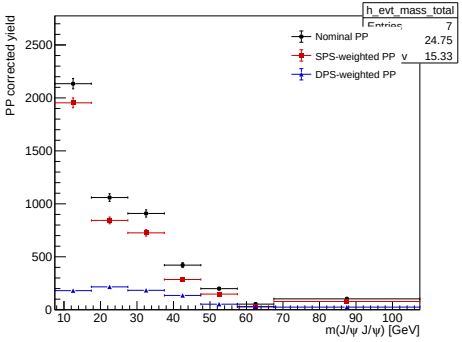
- Cell-wise signed SPS yield
- One negative cell retained
- Compatible with zero
- Correlations prevent a global-significance reading

Not shown

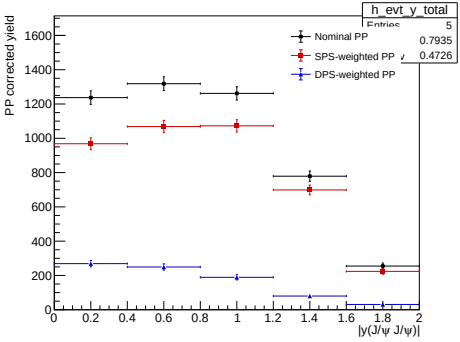
No truncation and no conversion to a probability map.

Backup: pair-kinematic PP-corrected yields

$m(J/\psi J/\psi)$ **yields**



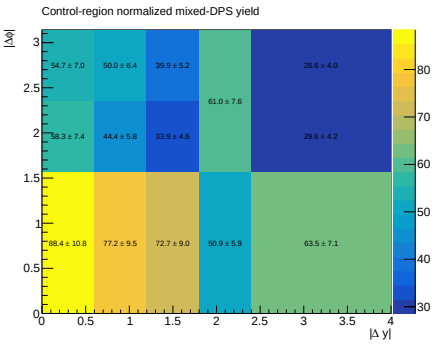
$|y(J/\psi J/\psi)|$ **yields**



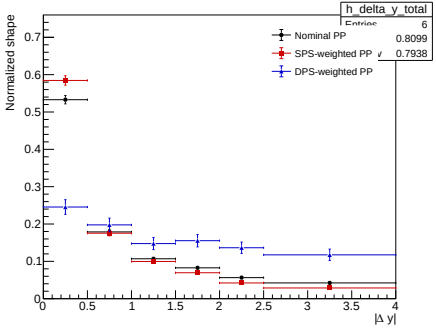
Projection fits preserve the central decomposition within correlated uncertainties.

Backup: generator-based DPS benchmark

DPS-MC normalized map



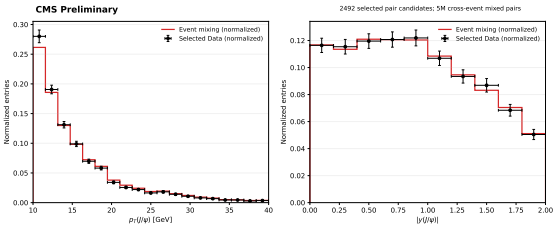
$|\Delta y|$ component shapes



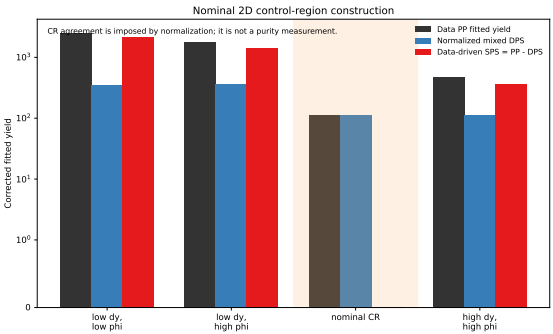
Benchmark direct $f_{\mathrm{DPS}} = 0.153752$ – generator-dependent alternative, not a data-driven uncertainty

Backup: earlier validation evidence

Previous mixed-candidate marginals

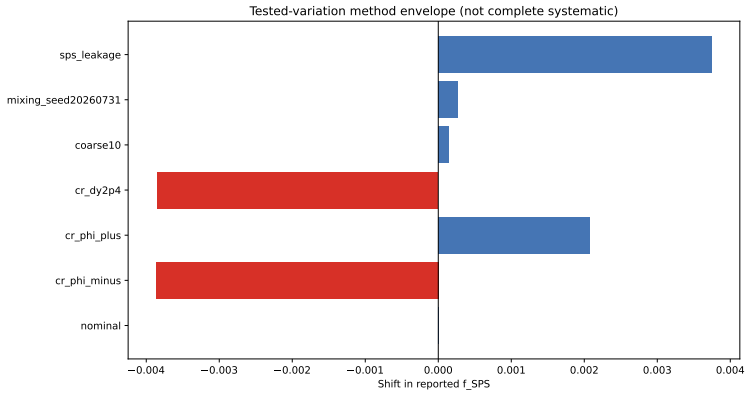


Previous CR boundary scan



Method-evolution context only; previous numerical definitions are superseded.

Backup: earlier uncertainty machinery



Still useful

- Tested-variation bookkeeping
- Included/missing-source separation
- Closure terminology

Superseded

The numerical fraction and statistical propagation shown here are not the current result.

Backup: bootstrap definition and failure policy

Stage	Recomputed per attempt	Success requirement
Resampling	2492 source-event draws	multiplicity closure
Single-J/ψ fits	two slots, two $\lvert y\rvert$ categories	valid calibration and sPlot
Mixed template	all accepted cross-event pairs	deterministic construction
PP decomposition	14 cell fits + CR scale	every fit accepted

Reported sample

500 successful replicas from 529 attempts; sample SD $= 0.056796$.

Conditional-selection caveat

Failures are fit- and sparsity-related; successful-only sampling may bias coverage.

The current 'stat' label must always travel with the 500/529 disclosure.